



## Modelling Financial Markets

Seminar 7: Value-at-Risk and Expected Shortfall — Hands-on



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## Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model); in event studies: Abnormal Return
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CF: Cornish–Fisher (expansion)
- ▣ CI: Confidence Interval
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ DM: Diebold–Mariano (test)
- ▣ EODHD: EOD Historical Data (market-data provider)
- ▣ ES: Expected Shortfall
- ▣ ETF: Exchange-Traded Fund
- ▣ EVT: Extreme Value Theory
- ▣ FHS: Filtered Historical Simulation
- ▣ FRTB: Fundamental Review of the Trading Book
- ▣ FZ0: Fissler–Ziegel loss, zero-homogeneous version
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GPD: Generalised Pareto Distribution
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HS: Historical Simulation
- ▣ LGD: Loss Given Default
- ▣ OMV: Österreichische Mineralölverwaltung (Austrian Mineral Oil Administration (OMV group))
- ▣ QML: Quasi-Maximum Likelihood
- ▣ RMSFE: Root Mean Squared Forecast Error
- ▣ SAV: Symmetric Absolute Value (CAViaR specification)
- ▣ SE: Standard Error

## Acronyms used in this material (2/2)

□ **US: United States**

## Today's Questions and Route

- ▣ Questions of the seminar
  - ▶ how large are tomorrow's tail losses?
  - ▶ how precisely can we tell?
- ▣ The seminar comes before Lecture 7: start with the section "What You Need for Today" (six slides); it gives every definition and formula the in-class tasks use
- ▣ Route in class
  - ▶ A1–A4: VaR and ES on paper, from the Normal distribution to a discrete default loss
  - ▶ B1, B6: historical VaR and ES on BET with bootstrap intervals; B3: tomorrow's filtered VaR for Bitcoin
  - ▶ C2: find the errors in an AI answer
- ▣ After the lecture, at home: B2 (an extreme-value tail), A5 and B5 (who carries the portfolio risk)
- ▣ What you produce: the A answers, one table and one figure per B exercise, the corrected C2 code, one interpretation sentence per exercise
- ▣ Notebook: [open the seminar notebook in Colab](#); each exercise has a notebook section with the same label
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## Exercise Map

| Exercise        | Question                                                                                          | Type                   | Worked example     |
|-----------------|---------------------------------------------------------------------------------------------------|------------------------|--------------------|
| A1, A2          | VaR and ES under the Normal and the Student- $t$ distributions                                    | Solved                 | –                  |
| A3              | VaR and ES of a discrete default loss                                                             | Solved                 | –                  |
| A4              | when VaR punishes diversification                                                                 | Proposed               | A3                 |
| B1              | historical VaR and ES on BET, with bootstrap intervals                                            | Solved                 | –                  |
| B6              | how precise is an ES estimate?                                                                    | Proposed               | B1                 |
| B3              | tomorrow's filtered VaR for Bitcoin                                                               | Proposed               | B1                 |
| C2              | what is wrong in an AI answer?                                                                    | Proposed               | B1                 |
| B2              | a GPD tail for BET (after the lecture)                                                            | Solved                 | –                  |
| A5, B5          | risk contributions of two assets and of a four-asset portfolio (after the lecture)                | Solved;<br>Proposed    | A5                 |
| Extension packs | A2, A4, A5, B1, B2, B3, B5 Extended; A6–A8, B4, B7; projects C1, C3 (optional, after the seminar) | Solved and<br>Proposed | named on each task |

- ▣ **[Solved]**: full solution in the slides and the notebook, a model to copy; **[Proposed]**: you solve it, the solution is discussed in class
- ▣ Part A: derivations on paper; Part B: estimation and inference on real data, each ending with an interpretation question; Part C: AI critique and open projects

## Data Used

| Series         | Price used                                                                    | Sample               | Calendar                        | Used in        |
|----------------|-------------------------------------------------------------------------------|----------------------|---------------------------------|----------------|
| BET            | daily close                                                                   | 2000–2026            | weekdays                        | B1, B2, B6, C2 |
| S&P 500        | daily close                                                                   | 2000–2026            | US trading days                 | B6             |
| Bitcoin        | daily close (USD)                                                             | 2014–2026            | every day                       | B3             |
| SPY, TLT, GLD  | adjusted close                                                                | 2014–2026            | common days with Bitcoin        | B5             |
| Gold; EUR/RON  | gold: daily close (USD per ounce); EUR/RON: BNR reference rate                | 2000–2026; 2005–2026 | weekdays; Romanian working days | B4             |
| BVB blue chips | Banca Transilvania, OMV Petrom and six other large BVB stocks, adjusted close | 2016–2026            | BVB trading days                | C1             |

- Daily prices of the course data, one series per asset (field used: close or adjusted close, as in the table); all samples end on 18 September 2026
- EUR/RON: the official BNR reference rate, not the EUR/RON series from EODHD (it contains erroneous quotes, Seminar 0, B6)
- Two or more assets: align the *prices* on common trading days first, then compute returns

## Setup: Run and Verify

```
import numpy as np, pandas as pd
URL = ('https://raw.githubusercontent.com/'
      'danpele/MFM/main/data/market/')
px = pd.read_csv(URL + 'BET.csv', index_col='date',
                 parse_dates=True)['close']
px = px.loc['2000-01-01':'2026-09-18']
px = px[px.index.dayofweek < 5]
r = 100 * np.log(px).diff().dropna()    # % log returns
L = -r                                  # losses
print(px.head(3)); print(len(r), r.index[-1])
```

- Code from the notebook section “Data”
- Expected output:  $n = 6\,670$  returns, last date 2026-09-18

| date       | close  | $r_t$ (%) |
|------------|--------|-----------|
| 2000-01-05 | 460.67 | –         |
| 2000-01-06 | 484.04 | 4.95      |
| 2000-01-07 | 526.34 | 8.38      |

- Last close: 32757.81 on 2026-09-18
- If your count or first value differs, check the series, the price column (close or adjusted close) and the weekday filter
- Other series: change only the series and the price column (slide “Data Used”)

## What You Need for Today (1/6): VaR and ES

- This seminar comes before the lecture: these six slides give only what A1–A4, B1, B3, B6 and C2 need; depth in [Chapter 7](#), sections “Value-at-Risk and Expected Shortfall”, “Coherent Risk Measures”, “Estimation Methods” and “Conditional VaR and ES” (the lecture after the seminar)
- Return  $X$ , loss  $L = -X$ ; the level  $\alpha$  is a **tail probability**: VaR 1%, ES 2.5%
- VaR: the loss exceeded with probability  $\alpha$ , i.e. minus the  $\alpha$ -quantile of the return
  - ▶  $\text{VaR}_\alpha = -q_\alpha(X) = q_{1-\alpha}(L)$ ,  $q_\alpha(X) = \inf\{x : P(X \leq x) > \alpha\}$
  - ▶  $q_\alpha(X)$ : the  $\alpha$ -quantile, the smallest  $x$  (infimum) at which the distribution function  $P(X \leq x)$  exceeds  $\alpha$
  - ▶ VaR says how often a loss is exceeded, not by how much
- ES: the average loss in the worst  $\alpha$  of outcomes; for a continuous loss
  - ▶  $\text{ES}_\alpha = \mathbb{E}[L \mid L \geq \text{VaR}_\alpha] \geq \text{VaR}_\alpha$ ;  $\mathbb{E}[\cdot \mid \cdot]$ : the mean computed only over the outcomes that satisfy the condition
  - ▶ Example: ten equally likely losses  $-2, -1, 0, 0, 1, 1, 2, 3, 5, 9$  at  $\alpha = 20\%$ : the worst 20% are  $\{9, 5\}$ , so  $\text{ES}_{20\%} = (9 + 5)/2 = 7$ ;  $P(L \geq 5) = 20\%$  is not above  $\alpha$ ,  $P(L \geq 3) = 30\%$  is, so  $\text{VaR}_{20\%} = 3$
- Regulation: the Basel market-risk rules replaced VaR 1% by ES 2.5% [Basel Committee \(2019\)](#)

## What You Need for Today (2/6): Parametric VaR and ES

- Normal returns:  $X = \mu + \sigma Z$ , with mean  $\mu$ , standard deviation  $\sigma$  and  $Z \sim N(0, 1)$ 
  - ▶  $\Phi, \varphi$ : distribution function and density of  $N(0, 1)$ ;  $z_\alpha = \Phi^{-1}(\alpha) < 0$ : its  $\alpha$ -quantile
  - ▶  $\text{VaR}_\alpha = -(\mu + \sigma z_\alpha)$ ,  $\text{ES}_\alpha = -\mu + \sigma \varphi(z_\alpha)/\alpha$  (A1 derives it): VaR is  $|z_\alpha|$  standard deviations below the mean, ES uses the larger factor  $\varphi(z_\alpha)/\alpha$
  - ▶  $z_{1\%} = -2.326$ ,  $\varphi(z_{1\%})/0.01 = 2.665$ ;  $z_{2.5\%} = -1.960$ ,  $\varphi(z_{2.5\%})/0.025 = 2.338$ ; with  $\mu = 0$ ,  $\sigma = 1\%$ :  $\text{VaR } 1\% = 2.33\%$ ,  $\text{ES } 2.5\% = 2.34\%$
- Student- $t$  returns:  $X = m + sT$ ,  $T \sim t_\nu$ 
  - ▶  $m$ : location;  $s > 0$ : scale;  $\nu$ : degrees of freedom, smaller  $\nu$  = heavier tails;  $T$  has variance  $\nu/(\nu - 2)$  for  $\nu > 2$
  - ▶ So  $X$  has variance  $s^2\nu/(\nu - 2)$ ; to match a standard deviation  $\sigma$  take  $s = \sigma\sqrt{(\nu - 2)/\nu}$  (A2)

## What You Need for Today (3/6): Coherence and Discrete Losses

- Coherent risk measure  $\rho$  [Artzner et al. \(1999\)](#): a rule that maps a position to the capital it needs, and is
  - ▶ monotone (larger losses, more capital), translation invariant (a sure loss  $c$  adds  $c$ ), positively homogeneous (double the position, double the capital)
  - ▶ and **subadditive**:  $\rho(A + B) \leq \rho(A) + \rho(B)$  for two positions  $A, B$ : merging positions never adds risk
  - ▶ ES is coherent; VaR can fail subadditivity when losses are discrete or very heavy-tailed (A4)
- With an atom (a default: a loss value with positive probability) the worst  $\alpha$  must be filled with probability mass [Acerbi & Tasche \(2002\)](#)(A3)
  - ▶  $ES_\alpha = \frac{1}{\alpha} (\mathbb{E}[L 1\{L > VaR_\alpha\}] + VaR_\alpha(\alpha - P(L > VaR_\alpha)))$
  - ▶  $1\{\cdot\}$ : indicator, 1 if the condition holds and 0 otherwise; first term: the losses strictly beyond VaR; second term: the missing tail probability, placed at VaR

## What You Need for Today (4/6): Historical Simulation and Its Precision

- Historical simulation (HS): no model for the distribution, only the past losses  $L_1, \dots, L_n$ 
  - ▶  $\widehat{\text{VaR}}_\alpha$ : the empirical  $(1 - \alpha)$  quantile of the losses;  $\widehat{\text{ES}}_\alpha$ : the mean of the  $k \approx n\alpha$  losses at or above it; the hat marks an estimate
  - ▶ The tail count  $n\alpha$  governs precision:  $n = 500$  days and  $\alpha = 1\%$  leave five losses for ES 1%
- Bootstrap Efron (1979): draw  $n$  losses with replacement, recompute the statistic, repeat  $B$  times; the 2.5% and 97.5% percentiles of the  $B$  values give a 95% interval
  - ▶ i.i.d. draws break volatility clusters (Chapter 1); the moving-block bootstrap Künsch (1989) draws blocks of, e.g., 20 consecutive days and keeps them, so the interval widens when crises cluster
- HS is unconditional: it describes the sample, a crisis-free window gives a low VaR even if today is volatile; that is why B3 filters the returns first

## What You Need for Today (5/6): Filtered VaR

- Filtered historical simulation (FHS) Barone-Adesi et al. (1999): fit an AR(1)-GARCH(1,1) (Chapter 5) and standardise the returns
  - ▶  $r_t = \mu_t + \sigma_t z_t$ ,  $\hat{z}_t = (r_t - \hat{\mu}_t) / \hat{\sigma}_t$
  - ▶  $\mu_t$ : conditional mean (AR(1));  $\sigma_t$ : conditional volatility (GARCH(1,1));  $z_t$ : i.i.d. shock with mean 0 and variance 1;  $\hat{z}_t$ : standardised residual
- Tomorrow's VaR: tomorrow's forecast scale times the residual quantile, minus the forecast mean
  - ▶  $\text{VaR}_{t+1} = -\hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \hat{\xi}_\alpha$ , with  $\hat{\xi}_\alpha = -q_\alpha(\hat{z})$ , minus the empirical  $\alpha$ -quantile of the residuals
  - ▶ ES: replace  $\hat{\xi}_\alpha$  by the mean of  $-\hat{z}$  beyond it
  - ▶ Example:  $\hat{\mu}_{t+1} = 0.1\%$ ,  $\hat{\sigma}_{t+1} = 3\%$ ,  $\hat{\xi}_{1\%} = 2.8$ :  $\text{VaR}_{1\%} = -0.1 + 3 \times 2.8 = 8.3\%$
- Out of sample: the forecast for day  $t + 1$  uses only data up to day  $t$

## What You Need for Today (6/6): Longer Horizons

- Horizon of  $h$  days: log losses add over days
  - ▶  $L_t^{(h)} = L_t + L_{t-1} + \dots + L_{t-h+1}$ : the loss over the  $h$  days ending on day  $t$
  - ▶ The  $h$ -day VaR is the quantile of these sums; consecutive sums share  $h - 1$  days (they overlap)
- Square-root-of-time rule:  $\text{VaR}^{(h)} = \sqrt{h} \text{VaR}^{(1)}$ , with  $\text{VaR}^{(h)}$  the  $h$ -day VaR
  - ▶ Exact only for i.i.d. zero-mean Normal daily losses: the variance of the sum is  $h$  times the daily variance, so the standard deviation grows like  $\sqrt{h}$
  - ▶ Autocorrelation and fat tails break this rule (C2)

## Notation and Conventions Used in Every Exercise

- ▣ Return  $X$  (in %), loss  $L = -X$ ; VaR and ES are **positive** numbers when they are losses
- ▣ Level  $\alpha$  = tail probability: VaR 1%, ES 2.5%;  $\text{VaR}_\alpha(X) = -\inf\{x : P(X \leq x) > \alpha\} = -q_\alpha(X)$ ,  
 $\text{ES}_\alpha(X) = -\frac{1}{\alpha} \int_0^\alpha q_u(X) du$
- ▣ Empirical convention (all B exercises)
  - ▶  $\widehat{\text{VaR}}_\alpha = \text{np.quantile}(L, 1 - \alpha)$ : linear interpolation at position  $1 + (n - 1)(1 - \alpha)$  of the sorted losses
  - ▶  $\widehat{\text{ES}}_\alpha = \text{mean of the } k \text{ losses } \geq \widehat{\text{VaR}}_\alpha, k \approx n\alpha$ : an approximation of the exact ES, equal to it when  $n\alpha$  is an integer
- ▣ Units: log losses in % of the position for one asset; simple returns for portfolios (they add across assets); percentage points (pp) for standard errors and interval widths
- ▣ Symbols that look alike:  $\alpha$  = tail probability,  $\alpha_1$  = GARCH coefficient;  $\xi$  = GPD shape,  $\hat{\xi}_\alpha$  = residual quantile (B3);  $w$  = money exposures (A5) or portfolio weights (B5)

## A1 [Solved]: Normal VaR and ES — Task

- ▣ **Question:** what are VaR and ES of a position whose daily return is Normal?
- ▣ **Data**
  - ▶  $X \sim N(\mu, \sigma^2)$ , daily log return in % of the position,  $\mu = 0.04\%$ ,  $\sigma = 1.2\%$ ; loss  $L = -X$  with density  $f$
  - ▶  $z_\alpha = \Phi^{-1}(\alpha) < 0$ ;  $\Phi$ ,  $\varphi$ : distribution and density functions of the standard Normal distribution
- ▣ **Tasks**
  1. Derive  $ES_\alpha = -\mu + \sigma\varphi(z_\alpha)/\alpha$  from  $ES_\alpha = -\mathbb{E}[X \mid X \leq q_\alpha(X)]$ .
  2. Compute  $z_\alpha$  and  $\varphi(z_\alpha)$  at  $\alpha = 1\%$  and  $\alpha = 2.5\%$ .
  3. Compute VaR and ES at both levels, in % of the position.
- ▣ **Report:** a four-cell table (VaR, ES  $\times$  1%, 2.5%)
- ▣ **Interpretation**
  - ▶ which of the four numbers are almost equal?
  - ▶ why does that matter for regulation?
- ▣ Notebook: section A1

## A1 [Solved]: Solution

- **1. Derivation.** Write  $X = \mu + \sigma Z$ ;  $X \leq q_\alpha$  is  $Z \leq z_\alpha$ , of probability  $\alpha$

► Since  $\varphi'(u) = -u\varphi(u)$ :

$$\int_{-\infty}^z u\varphi(u) du = [-\varphi(u)]_{-\infty}^z = -\varphi(z)$$

►  $\mathbb{E}[X \mid X \leq q_\alpha] = \mu + \frac{\sigma}{\alpha} \int_{-\infty}^{z_\alpha} u\varphi(u) du = \mu - \sigma \frac{\varphi(z_\alpha)}{\alpha}$ ;  
change the sign

- **2–3.**  $\text{VaR}_\alpha = -(\mu + \sigma z_\alpha)$ , e.g.  
 $-(0.04 + 1.2 \cdot (-2.326)) = 2.75$ ;  
 $\text{ES}_{1\%} = -0.04 + 1.2 \cdot 2.665 = 3.16$

| $\alpha$ | $z_\alpha$ | $\varphi(z_\alpha)$ | $\varphi/\alpha$ | $\text{VaR}_\alpha$ | $\text{ES}_\alpha$ |
|----------|------------|---------------------|------------------|---------------------|--------------------|
| 1%       | -2.326     | 0.0267              | 2.665            | 2.75                | 3.16               |
| 2.5%     | -1.960     | 0.0584              | 2.338            | 2.31                | 2.77               |

- VaR and ES in % of the position; for a position of 1 million currency units: VaR 1%  $\approx 27\,516$ , ES 1%  $\approx 31\,583$
- Interpretation: ES 2.5% (2.77%)  $\approx$  VaR 1% (2.75%): under the Normal distribution the switch to ES 2.5% keeps the capital, which is why FRTB chose this level

## A2 [Solved]: Student- $t$ ES — Task

▣ **Question:** how much do heavy tails change VaR and ES when the standard deviation is the same as in A1?

▣ **Data**

►  $X = m + sT$ ,  $T \sim t_\nu$  with density  $g_\nu$ ,  $t_\alpha = t_\nu^{-1}(\alpha) < 0$ ;  $\nu = 4$ ,  $m = 0.04\%$ , standard deviation 1.2%

► Tail integral (proof in Extension Pack 1):  $\int_{-\infty}^t x g_\nu(x) dx = -g_\nu(t) \frac{\nu + t^2}{\nu - 1}$  for  $\nu > 1$

▣ **Tasks**

1. Use the tail integral to write  $ES_\alpha = -m + s \frac{g_\nu(t_\alpha)}{\alpha} \frac{\nu + t_\alpha^2}{\nu - 1}$ .
2. Find the scale  $s$  that gives a standard deviation of 1.2%, using  $\text{Var}(T) = \nu/(\nu - 2)$  (finite only for  $\nu > 2$ ).
3. Compute VaR and ES at 1% and 2.5% and put them next to A1.

▣ **Report:** the A1 table with a Student- $t$  column

▣ **Interpretation**

- at which level do the two models differ most?
- why does ES see the difference first?

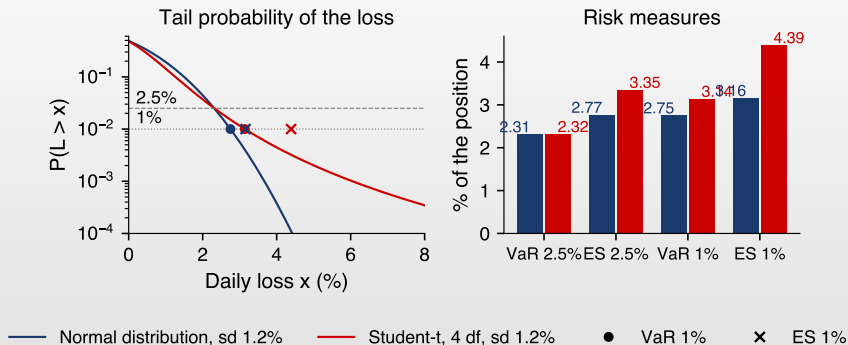
▣ Common mistake: using  $\sigma = 1.2$  instead of  $s$  as the scale

## A2 [Solved]: Solution

- 1. Divide the tail integral by  $\alpha$  and rescale with  $X = m + sT$ , as in A1
- 2.  $s^2 \cdot 4/2 = 1.2^2$ , so  $s = 1.2\sqrt{2/4} = 0.849$
- 3. At 1%:  $t_{1\%} = -3.747$ ,  $g_4(t_{1\%}) = 0.0087$ , ES factor  $\frac{0.0087}{0.01} \cdot \frac{4 + 14.04}{3} = 5.221$ 
  - $\text{VaR}_{1\%} = -(0.04 + 0.849 \cdot (-3.747)) = 3.14$ ;  $\text{ES}_{1\%} = -0.04 + 0.849 \cdot 5.221 = 4.39$

|        | $\text{VaR}_{2.5\%}$ | $\text{ES}_{2.5\%}$ | $\text{VaR}_{1\%}$ | $\text{ES}_{1\%}$ |
|--------|----------------------|---------------------|--------------------|-------------------|
| Normal | 2.31                 | 2.77                | 2.75               | 3.16              |
| $t_4$  | 2.32                 | 3.35                | 3.14               | 4.39              |

## A2 [Solved]: Solution — Chart



- Interpretation: VaR 2.5% is the same in both models; VaR 1% and both ES are much higher under  $t_4$ : heavy tails show only deep in the tail, and ES sees them first

## A3 [Solved]: VaR and ES of a Discrete Loss — Task

- **Question:** what are VaR and ES when the loss has an atom (a default)?
- **Data:** a bond loses 60 monetary units (loss given default, LGD) with probability  $p = 3\%$  and 0 otherwise
- **Tasks**
  1. Compute VaR 5% and VaR 2.5% from  $\text{VaR}_\alpha = -\inf\{x : P(X \leq x) > \alpha\}$ .
  2. Compute ES 5% and ES 2.5% with  $\text{ES}_\alpha = \frac{1}{\alpha} (\mathbb{E}[L1\{L > \text{VaR}_\alpha\}] + \text{VaR}_\alpha(\alpha - P(L > \text{VaR}_\alpha)))$  Acerbi & Tasche (2002).
  3. Explain why  $\text{ES}_{5\%}$  differs from  $\mathbb{E}[L \mid L \geq \text{VaR}_{5\%}]$ .
- **Report:** four numbers and one sentence

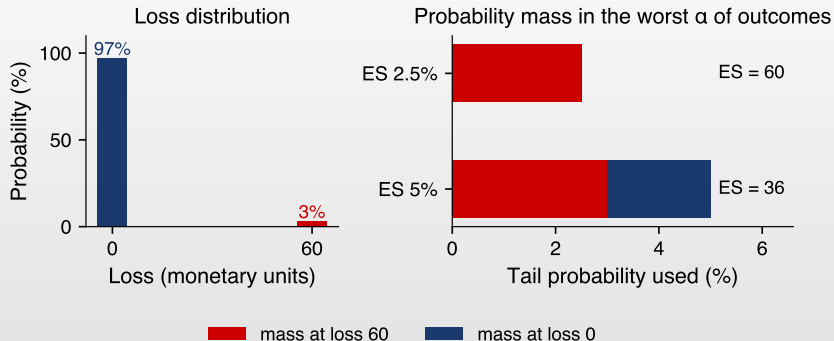
| Loss $L$          | 0    | 60 |
|-------------------|------|----|
| $P(L)$            | 97%  | 3% |
| $P(L \geq \cdot)$ | 100% | 3% |

- The worst  $\alpha$  of outcomes must be filled with probability mass; at  $\alpha = 5\%$  the 3% default mass is not enough

## A3 [Solved]: Solution

- 1.  $P(X \leq -60) = 3\% \leq 5\%$ ,  $P(X \leq 0) = 100\% > 5\%$ :  $q_{5\%} = 0$ ,  $\text{VaR}_{5\%} = 0$ ;  $P(X \leq -60) = 3\% > 2.5\%$ :  $\text{VaR}_{2.5\%} = 60$
- 2.  $\text{ES}_{5\%} = \frac{60 \cdot 0.03 + 0 \cdot (0.05 - 0.03)}{0.05} = 36$ ;  $\text{ES}_{2.5\%} = \frac{0 + 60 \cdot (0.025 - 0)}{0.025} = 60$
- 3.  $\mathbb{E}[L \mid L \geq 0] = 1.8$  averages over **all** outcomes; ES 5% uses the 3% default mass plus 2% of the zero-loss atom (chart)
- Interpretation: this fractional mass is the exact version of the tail average used in B1 with  $k \approx n\alpha$  losses

## A3 [Solved]: Solution — Chart



## A4 [Proposed]: When VaR Punishes Diversification — Task

- **Question:** can a VaR limit make a diversified portfolio look riskier than its parts?
- **Data:** two independent bonds A and B; each loses 100 with probability  $p = 0.7\%$  (the lecture used 0.9%) and 0 otherwise; portfolio A + B [Artzner et al. \(1999\)](#)
- **Tasks**
  1. Write the distribution of the portfolio loss: the values 0, 100, 200 and their probabilities.
  2. Compute VaR 1% and ES 1% of one bond and of the portfolio with the discrete formula of A3.
  3. Show that VaR 1% is not subadditive here while ES 1% is.
  4. Repeat steps 2–3 at  $\alpha = 1.5\%$ .
- **Report:** a table of VaR and ES for A, B and A + B at both levels
- **Interpretation:** what would a VaR-based limit reward?
- **Worked example to follow:** A3

## B1 [Solved]: Historical VaR and ES on BET — Task

### ▣ Questions

- ▶ how large are the tail losses of the Romanian stock market?
- ▶ how precise are the estimates?

### ▣ Data

- ▶ BET index, daily closes 2000-01-05 – 2026-09-18, weekdays only;  $n = 6\,670$  daily log returns in %,  $r_t = 100 (\ln P_t - \ln P_{t-1})$ , loss  $L_t = -r_t$
- ▶ Two samples: the full period and the last 500 returns (from 2024-09-12, shaded in the chart)

### ▣ Tasks

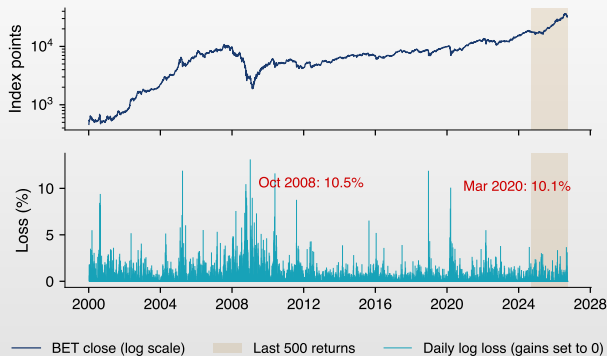
1. Compute historical VaR and ES at 1% and 2.5% on both samples, and count the losses beyond VaR 1%.
2. Build a 95% percentile bootstrap interval for VaR 1% with 2 000 i.i.d. resamples, seed 42 [Efron \(1979\)](#).
3. Repeat step 2 with a moving-block bootstrap: 20-day blocks drawn with replacement, concatenated and cut to length  $n$  [Künsch \(1989\)](#).

### ▣ Report: a table of the eight VaR/ES numbers and the two intervals per sample

### ▣ Interpretation

- ▶ which estimate would you give a risk committee?
- ▶ why is the block interval wider on the full sample?

## B1 [Solved]: Historical VaR and ES on BET — Data



- Look first: the largest losses (October 2008: 10.45%; March 2020: 10.08%) come in clusters

## B1 [Solved]: Solution (1/3) — Step by Step

- **1. Last 500 days.** Sort the losses; VaR 1% sits at position  $1 + 499 \cdot 0.99 = 495.01$ ;  $L_{(i)}$ : the  $i$ -th smallest loss

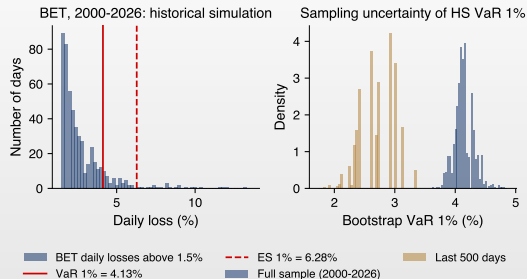
▶  $\widehat{\text{VaR}}_{1\%} = L_{(495)} + 0.01 (L_{(496)} - L_{(495)}) = 2.710 + 0.01 (2.925 - 2.710) = 2.712\%$

▶ The five losses beyond it: 3.68, 3.35, 3.12, 3.03, 2.93: their mean is  $\widehat{\text{ES}}_{1\%} = 3.22\%$

|               | $n$   | $\text{VaR}_{1\%}$ | $\text{ES}_{1\%}$ | $\text{VaR}_{2.5\%}$ | $\text{ES}_{2.5\%}$ | beyond VaR 1% |
|---------------|-------|--------------------|-------------------|----------------------|---------------------|---------------|
| 2000–2026     | 6 670 | 4.13               | 6.28              | 2.80                 | 4.57                | 67            |
| Last 500 days | 500   | 2.71               | 3.22              | 2.05                 | 2.69                | 5             |

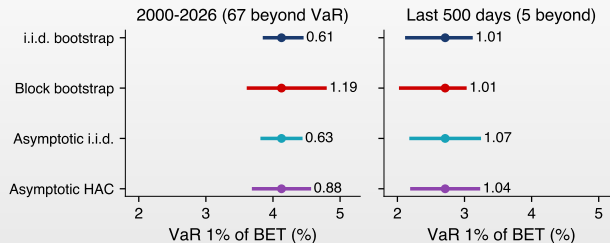
- The 500-day ES 1% averages only five numbers: report it with that count
- Both are unconditional summaries: the full sample describes 26 years, the last 500 days a calm period; tomorrow's risk needs a conditional forecast (B3)

## B1 [Solved]: Solution (2/3) — The Tail and the Bootstrap



- Left: BET losses above 1.5% with historical VaR 1% and ES 1%; right: the 2000 i.i.d. bootstrap values of VaR 1% for each sample
- Each resample recomputes both the quantile and which losses lie beyond it; the 2.5% and 97.5% percentiles of the 2000 values give the interval
- The two bootstrap distributions do not overlap: the recent market is calmer than the 26-year average

## B1 [Solved]: Solution (3/3) — Intervals



Dot: historical VaR 1%; bar: 95% interval; number: interval width in percentage points

- 2. i.i.d. bootstrap: full [3.85, 4.46], last 500 days [2.11, 3.12]
- 3. Block bootstrap: [3.61, 4.80] and [2.02, 3.03]

- Interpretation: exceedances cluster in 2008 and 2020, so blocks that keep a crisis together widen the full-sample interval by about half (in B1 Extended: asymptotic and HAC); the calm last 500 days show no such effect

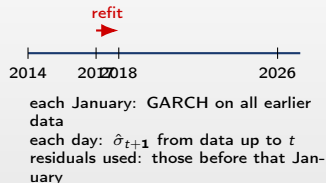
## B6 [Proposed]: How Precise Is an ES Estimate? — Task

- **Question:** does a 500-day historical ES 2.5% pin the tail risk down to within  $\pm 0.3$  pp [Pele & Mazurencu-Marinescu-Pele \(2026\)](#)?
- **Data**
  - ▶ S&P 500 (daily close) and BET, daily log losses in %, 2000–2026; samples: full and last 500 days
  - ▶ Use the resampling function of B1 (2 000 resamples, seed 42, 20-day blocks) with ES 2.5% as the statistic
- **Tasks**
  1. Compute ES 2.5%, the tail count  $n\alpha$  and the number  $k$  of losses that the tail mean actually averages.
  2. Draw one i.i.d. and one block resample of the S&P 500 last 500 days, and recompute VaR 2.5%, the tail membership and ES on each.
  3. Build the i.i.d. and the block 95% intervals for all four samples and compare their widths with 0.6 pp ( $\pm 0.3$ ).
- **Report:** a table ( $n\alpha$ ,  $k$ , ES, two intervals) and an interval chart
- **Interpretation**
  - ▶ which samples meet the  $\pm 0.3$  pp target?
  - ▶ what governs the width?
- **Worked example to follow:** B1

## B3 [Proposed]: Tomorrow's Filtered VaR for Bitcoin — Task

- **Question:** Bitcoin's VaR 1% for 19 September 2026: filtered or historical?
- **Data**
  - ▶ Bitcoin (USD), daily close, 17 September 2014 – 18 September 2026, 7 days a week; daily log returns in %
  - ▶ AR(1)-GARCH(1,1) by quasi-maximum likelihood (QML):  $r_t = \mu_t + \sigma_t z_t$  [Barone-Adesi et al. \(1999\)](#)
- **Tasks**
  1. Estimate the model on the full sample and forecast  $\hat{\mu}_{t+1}$ ,  $\hat{\sigma}_{t+1}$ .
  2. Compute  $\hat{\xi}_{1\%} = -q_{1\%}(\hat{z})$  from the standardised residuals and the filtered VaR 1% and ES 1%.
  3. Compare with historical VaR and ES on the last 500 days and on the full sample.
  4. With the forecasts of the timeline (right), count the days of the last two years with a loss above the FHS VaR and above a rolling 500-day HS VaR.
- **Report:** three VaR/ES pairs, two exceedance counts
- **Worked example to follow:** B1

### Forecast timeline (step 4)



- No forecast uses information from its own day or later: a full-sample threshold is not an out-of-sample forecast
- **Interpretation**
  - ▶ why do the three VaRs differ?
  - ▶ which method misses more often?

## C2 [Proposed]: Critique an AI Answer — the Answer

- **Prompt:** "px holds daily BET closes, 2000–2026. Give me the one-day historical VaR 1% and ES 2.5% as positive log losses in %, and the 10-day VaR 1%."
- **Answer:** the code below, which runs without an error message

```
1  import numpy as np
2  r = 100 * np.log(px).diff().dropna()           # daily log returns, %
3  loss = -r                                       # log loss, %
4  var1 = np.percentile(r, 99)                   # VaR 1%
5  es25 = loss[loss >= np.percentile(loss, 99)].mean() # ES 2.5%
6  var10 = 10 * var1                             # 10-day VaR 1%
```

- **Conclusion of the AI:** "VaR 1% = 3.95%, ES 2.5% = 6.28%, 10-day VaR 1% = 39.46%."

## C2 [Proposed]: Critique an AI Answer — Task

### ▣ Questions

- ▶ which lines of the AI answer are wrong?
- ▶ what are the correct numbers?

▣ **Data:** the BET data of B1; the answer contains **three** mistakes typical for VaR and ES, everything else is correct

### ▣ Tasks

1. For each mistake, name the line number and explain in one sentence why it is wrong, using  $\text{VaR}_\alpha = -q_\alpha(r) = q_{1-\alpha}(L)$ .
2. Say how you would detect it: a count of observations, a number from B1, or a slide of the lecture.
3. Write the corrected lines, including the 10-day VaR from overlapping 10-day losses, and recompute the three numbers.
4. Check the corrected ES 2.5%: how many losses does it average? How many should it average?

▣ **Report:** an error table (line, error, detection, correction) and the corrected numbers; log each error you actually found in `AI_ERRORS.md`

▣ **Interpretation:** why does the code run without an error message although the answer is wrong?

▣ **Worked example to follow:** B1

## After the Lecture: What Comes Next

- These three exercises use tools that Lecture 7 develops: extreme value theory (B2) and Euler risk contributions (A5, B5); solve them after the lecture
- Depth: [Chapter 7](#), sections “Extreme Value Theory” and “Portfolio VaR and Risk Contributions”
- B2 and A5 are solved: read them as worked examples; B5 is the proposed transfer exercise (worked example: A5)

## B2 [Solved]: A GPD Tail for BET — Task

□ **Question:** what does an extreme-value tail add beyond historical simulation, deep in the tail?

□ **Data**

- ▶ BET index, daily closes 2000-01-05 – 2026-09-18, weekdays only;  $n = 6670$  daily log returns in %,  $r_t = 100 (\ln P_t - \ln P_{t-1})$ , loss  $L_t = -r_t$
- ▶ Peaks over threshold: excesses  $Y = L - u$  above a high threshold  $u$  follow a generalised Pareto distribution (GPD) with shape  $\xi$  ( $\xi > 0$ : power-law, heavy tail;  $\xi = 0$ : exponential-type tail, as for the Normal distribution) and scale  $\beta > 0$  [McNeil, Frey & Embrechts \(2015\)](#);  $n$  = sample size,  $N_u$  = number of exceedances

□ **Tasks**

1. Set  $u$  = the empirical 95% quantile of  $L$  and form the excesses  $L - u$  of the losses above it.
2. Fit the GPD by maximum likelihood with location fixed at 0: `scipy.stats.genpareto.fit(excesses, floc=0)`.
3. Compute the i.i.d. standard errors  $\text{SE}(\hat{\xi}) = (1 + \hat{\xi})/\sqrt{N_u}$  and  $\text{SE}(\hat{\beta}) = \hat{\beta}\sqrt{2(1 + \hat{\xi})/N_u}$ .
4. Compute GPD VaR and ES at 1%, 0.5% and 0.1% with  $\text{VaR}_\alpha = u + \frac{\beta}{\xi} [(\frac{n}{N_u}\alpha)^{-\xi} - 1]$ ,  $\text{ES}_\alpha = \frac{\text{VaR}_\alpha + \beta - \xi u}{1 - \xi}$ , and compare with historical simulation.
5. Refit for thresholds leaving 10% down to 1.5% of the losses above them and plot  $\hat{\xi}$  with 95% bands.

□ **Report:** the fit with standard errors, a six-row VaR/ES table with tail counts, and the stability chart

□ **Interpretation**

- ▶ is the BET tail heavier than the Normal distribution?
- ▶ how sure are we?

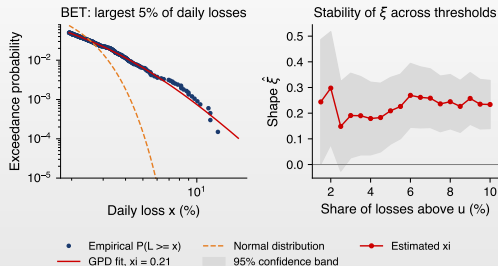
## B2 [Solved]: Solution (1/2) — Fit and Table

- 1–2.  $u = 1.93\%$  leaves  $N_u = 334$  of  $n = 6\,670$  losses above it;  $\hat{\xi} = 0.209$ ,  $\hat{\beta} = 1.20$
- 3.  $SE(\hat{\xi}) = (1 + 0.209)/\sqrt{334} = 0.066$ ,  $SE(\hat{\beta}) = 0.10$ ;  $\hat{\xi}/SE = 3.2$
- 4. VaR 1%:  $N_u/n = 0.050$ ,  $(0.01/0.050)^{-0.209} = 1.401$ 
  - ▶  $VaR_{1\%} = 1.93 + \frac{1.20}{0.209}(1.401 - 1) = 4.22\%$

|      | VaR GPD | VaR hist. | ES GPD | ES hist. |
|------|---------|-----------|--------|----------|
| 1%   | 4.22    | 4.13      | 6.34   | 6.28     |
| 0.5% | 5.47    | 5.38      | 7.92   | 7.86     |
| 0.1% | 9.18    | 9.50      | 12.61  | 11.25    |

- Losses beyond historical VaR: 67 at 1%, 34 at 0.5%, 7 at 0.1%
- The two methods agree down to 0.5%; at 0.1% historical ES averages 7 numbers, while the GPD uses all 334 excesses through a fitted tail

## B2 [Solved]: Solution (2/2) — Diagnostics



- Left: the GPD follows the empirical tail, the Normal tail falls far below it; right:  $\hat{\xi}$  ranges from 0.15 to 0.30 across thresholds, while VaR 0.1% moves only between 9.04% and 9.51%
- Interpretation: the tail is heavier than the Normal one, but the i.i.d. SE assumes independent exceedances; BET losses cluster, so the honest uncertainty is larger (B2 Extended: extremal index)

## A5 [Solved]: Risk Contributions of Two Assets — Task

□ **Question:** does each asset carry risk in proportion to the money invested in it?

□ **Data**

- ▶ Money exposures  $w = (600\,000, 400\,000)$  EUR in equity and bonds; daily simple returns  $R$  in decimals, standard deviations  $\sigma_1 = 0.012$ ,  $\sigma_2 = 0.009$ , correlation  $\rho = -0.2$
- ▶  $\Sigma = \begin{pmatrix} 0.000144 & -0.0000216 \\ -0.0000216 & 0.000081 \end{pmatrix}$ ; ( $\Sigma$ : covariance matrix) jointly Normal returns, mean ignored,  $\alpha = 1\%$ ,  $z_{99\%} = 2.326$  (the 99% standard Normal quantile)

□ **Tasks**

1. Compute the portfolio standard deviation  $\sigma_p = \sqrt{w^\top \Sigma w}$  in EUR and the portfolio VaR  $1\% = z_{99\%} \sigma_p$ .
2. Compute  $\Sigma w$ , the marginal VaR  $z_{99\%}(\Sigma w)_i / \sigma_p$  per EUR (the change in VaR per extra EUR in asset  $i$ ) and the component VaR  $C_i = w_i \times \text{marginal VaR}$ ; check  $C_1 + C_2 = \text{VaR}$  [Tasche \(2008\)](#).
3. Compute the diversification benefit: the sum of stand-alone VaRs minus the portfolio VaR.
4. Under the Normal distribution, rescale each  $C_i$  by  $\frac{\varphi(z_{1\%})/0.01}{z_{99\%}}$  to get the ES 1% contributions.

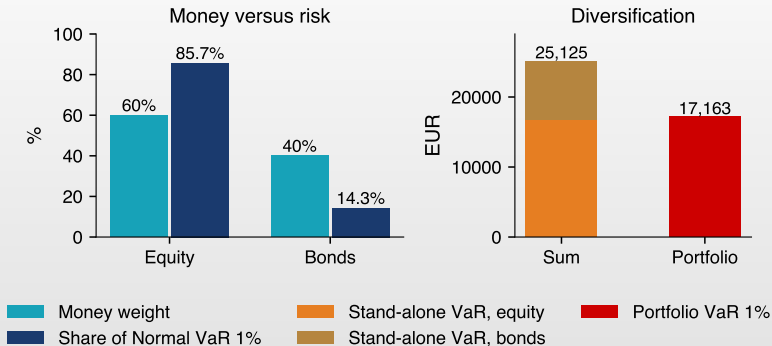
□ **Report:** a table of  $C_i$ ,  $C_i^{ES}$  and shares, and the diversification benefit in EUR

□ **Interpretation:** do the bonds carry risk in proportion to their 40% money weight?

## A5 [Solved]: Solution

- 1.  $\sigma_p^2 = 7\,200^2 + 3\,600^2 + 2(-0.2)(7\,200)(3\,600) = 54\,432\,000$ ,  $\sigma_p = 7\,378$  EUR,  $\text{VaR}_{1\%} = 2.326 \cdot 7\,378 = 17\,163$  EUR
- 2.  $\Sigma w = (77.76, 19.44)$ ; marginal VaR per EUR (0.0245, 0.0061);  $C_1 = 14\,711$  (85.7%),  $C_2 = 2\,452$  (14.3%), sum 17 163 EUR
- 3. Stand-alone  $16\,750 + 8\,375 = 25\,125$  EUR vs 17 163 EUR
- 4. Factor 1.146:  $C_1^{ES} = 16\,854$ ,  $C_2^{ES} = 2\,809$ , total 19 663 EUR = ES 1%; same shares as VaR under the Normal distribution

## A5 [Solved]: Solution — Chart



□ Interpretation: the bonds hold 40% of the money but 14.3% of the risk: the negative correlation makes them a partial hedge

## B5 [Proposed]: Risk Contributions of a Four-Asset Portfolio — Task

- **Question:** in an equally weighted portfolio of SPY, TLT, GLD and Bitcoin, who carries the risk?
- **Data**
  - ▶ SPY, TLT, GLD (adjusted closes) and Bitcoin (USD closes), 18 September 2014 – 18 September 2026; prices aligned on the 3017 common days, then daily simple returns  $R$  in %
  - ▶ Fractional weights  $w_i = 25\%$ , rebalanced daily;  $\Sigma$  = sample covariance of  $R$  (in  $\%^2$ ); Normal VaR 1% with the mean ignored; historical ES 2.5%
- **Tasks**
  1. Compute the Normal VaR 1%, the marginal VaR and the component VaR of each asset, as in A5.
  2. Compute the historical ES 2.5% of the portfolio loss  $L_p = -w^\top R$  and the contributions  $\mathbb{E}[-w_i R_i \mid L_p \geq \text{VaR}_{2.5\%}]$  as averages over the tail days.
  3. Compute the undiversified VaR  $\sum_i z_{99\%} w_i \sigma_i$ .
- **Report:** a table of weights, contributions and shares, and a chart of money weights vs risk shares
- **Interpretation:** is equal money equal risk? (comparing Normal VaR 1% with historical ES 2.5% changes both the model and the level)
- **Worked example to follow:** A5

## Exit Check and Team Project

### Exit check: can you say why

- a 500-day ES 1% should be reported with the number 5?
- $\sqrt{10}$  scaling is too low for BET?
- a 25% position can carry 80% of the risk (B5, after the lecture)?
- portfolio losses need simple, not log returns?

### Team project

- Choose one of C1 or C3 (Extension Pack 4), with the common rubric
- Log AI use in AI\_USE.md and every error you actually caught in AI\_ERRORS.md (no quota)

## Extension Packs: What Each One Needs

| Pack                     | Exercises                                                                 | Prerequisites                 |
|--------------------------|---------------------------------------------------------------------------|-------------------------------|
| 1. Tail algebra          | A2 Extended [Solved], A4 Extended, A6, A7, A5<br>Extended [Solved], A8    | A1–A5                         |
| 2. Inference             | B1 Extended [Solved], B3 Extended, B4, B5<br>Extended                     | B1, B3, B5                    |
| 3. Extremes and dynamics | B2 Extended [Solved], B7                                                  | B2, B3                        |
| 4. Projects              | C1 (ES capital, BVB vs S&P 500), C3 (deep<br>quantile regression); rubric | the lecture case study, B1–B3 |

- Each exercise has a notebook section with the same label and names its worked example; do them in order within a pack

## A2 Extended [Solved]: The Student- $t$ Tail Integral — Task and Solution

- **Question:** where does the Student- $t$  ES factor of A2 come from?
- **Data:**  $g_\nu(x) = c_\nu(1 + x^2/\nu)^{-(\nu+1)/2}$ , the  $t_\nu$  density;  $c_\nu$ : the constant that makes it integrate to 1
- **Tasks**
  1. Differentiate  $(1 + x^2/\nu)^{-(\nu+1)/2}$ .
  2. Deduce  $\int_{-\infty}^t x g_\nu(x) dx = -g_\nu(t) \frac{\nu+t^2}{\nu-1}$  and state where  $\nu > 1$  is needed.
- **Report:** the two-line proof

### Solution

- 1.  $\frac{d}{dx}(1 + x^2/\nu)^{-(\nu+1)/2} = -\frac{\nu+1}{\nu} x (1 + x^2/\nu)^{-(\nu+1)/2}$ , so  $x g_\nu(x) = -\frac{\nu c_\nu}{\nu-1} \frac{d}{dx}(1 + x^2/\nu)^{-(\nu+1)/2}$
- 2. Integrating from  $-\infty$  to  $t$ :  $-\frac{\nu}{\nu-1} g_\nu(t)(1 + t^2/\nu) = -g_\nu(t) \frac{\nu+t^2}{\nu-1}$ ; the term at  $-\infty$  vanishes only if  $\nu > 1$
- ES needs  $\nu > 1$ ; matching a variance needs  $\nu > 2$

## A4 Extended [Proposed]: For Which Levels Does VaR Fail? — Task

- ▣ **Question:** for which tail probabilities  $\alpha$  is VaR not subadditive in A4?
- ▣ **Data:** the A4 bonds,  $p = 0.7\%$ , and the definition  $\text{VaR}_\alpha = -\inf\{x : P(X \leq x) > \alpha\}$
- ▣ **Tasks**
  1. Write  $\text{VaR}_\alpha(A)$  and  $\text{VaR}_\alpha(A + B)$  as step functions of  $\alpha$ .
  2. Find the interval of  $\alpha$  where  $\text{VaR}_\alpha(A + B) > 2\text{VaR}_\alpha(A)$  and explain both endpoints.
- ▣ **Report:** the interval and one sentence per endpoint
- ▣ **Worked example to follow:** A4

## A6 [Proposed]: When Is Cornish–Fisher a Valid Quantile? — Task

□ **Question:** for which excess kurtosis is the Cornish–Fisher map still a quantile function?

□ **Data**

▶  $\mu = 0$ ,  $\sigma = 1.2\%$ , skewness  $S = -0.5$ , excess kurtosis  $K$  [Cornish & Fisher \(1938\)](#)

▶  $\tilde{z}(z) = z + \frac{(z^2-1)S}{6} + \frac{(z^3-3z)K}{24} - \frac{(2z^3-5z)S^2}{36}$ ,  $\text{VaR}_\alpha = -\sigma\tilde{z}(z_\alpha)$

□ **Tasks**

1. With  $K = 3$ , compute the three correction terms at  $z_{1\%}$ ,  $\tilde{z}_{1\%}$  and the Cornish–Fisher VaR 1%.
2. Write  $d\tilde{z}/dz = Az^2 + Bz + C$  and locate the vertex of this parabola.
3. Find the largest  $K$  for which  $\tilde{z}$  is increasing on  $[z_{0.1\%}, 0]$ , and decide whether  $K = 10$  is admissible.

□ **Report:** the VaR, the bound on  $K$  and a plot of  $\tilde{z}$  and  $d\tilde{z}/dz$  for  $K = 3$  and  $K = 10$

□ **Interpretation:** does a valid quantile map also give an accurate VaR?

□ **Worked example to follow:** A1

## A7 [Proposed]: ES as a Minimum, and Subadditivity — Task

- **Question:** why can ES never reward splitting a portfolio?
- **Data**
  - ▶ Integrable loss  $L$ ;  $(x)_+ = \max(x, 0)$ ; objective  $G_\alpha(v) = v + \mathbb{E}[(L - v)_+]/\alpha$  Rockafellar & Uryasev (2000)
  - ▶ Numbers: the A4 portfolio (two bonds,  $p = 0.7\%$ ) at  $\alpha = 1\%$
- **Tasks**
  1. For the A4 portfolio, compute the slopes of  $G$  on  $(0, 100)$  and  $(100, 200)$ , the minimiser and the minimum.
  2. Show that the right derivative of  $G$  is  $1 - P(L > v)/\alpha$  and deduce that  $\text{VaR}_\alpha$  minimises  $G$ , with minimum  $\text{ES}_\alpha$ .
  3. Deduce  $\text{ES}_\alpha(L_1 + L_2) \leq \text{ES}_\alpha(L_1) + \text{ES}_\alpha(L_2)$  from  $(a + b)_+ \leq a_+ + b_+$ .
- **Report:** the three numbers of step 1 and two short proofs
- **Worked example to follow:** A3, A4

## A5 Extended [Solved]: ES Contributions from the Minimum — Task and Solution

- **Question:** why are the historical ES contributions of B5 averages over the tail days?
- **Data:**  $ES_\alpha(w) = \min_v G(w, v)$ ,  
 $G(w, v) = v + \mathbb{E}[(L_p - v)_+]/\alpha$ ,  $L_p = -w^\top R$  (A7)
- **Tasks**
  1. Use the envelope argument at  $v^* = \text{VaR}_\alpha$  to compute  $\partial ES_\alpha / \partial w_i$ .
  2. Show that  $w_i \partial ES_\alpha / \partial w_i = \mathbb{E}[-w_i R_i \mid L_p \geq \text{VaR}_\alpha]$  for a continuous  $L_p$ .
- **Report:** a short proof
- **Worked example to follow:** A7

### Solution

- $\partial G / \partial v = 0$  at  $v^*$ , so  
 $\partial ES_\alpha / \partial w_i = \partial G / \partial w_i(w, v^*) = \mathbb{E}[-R_i \mathbf{1}\{L_p > v^*\}]/\alpha$
- For continuous  $L_p$ ,  $P(L_p > v^*) = \alpha$ : this is  $\mathbb{E}[-R_i \mid L_p \geq \text{VaR}_\alpha]$ ; summing  $w_i \times$  it gives  $ES_\alpha$  (Euler)
- Historical estimate: average  $-w_i R_i$  over the tail days (B5); under the Normal distribution it equals the VaR rescaling of A5

## A8 [Proposed]: The Sampling Variance of Historical ES — Task

- ▣ **Question:** how imprecise is a 500-day historical ES 2.5%, even for Normal losses?
- ▣ **Data**
  - ▶  $n$  i.i.d. losses, continuous, positive density at  $\text{VaR}_\alpha$ ,  $\mathbb{E}[L^2] < \infty$ ;  $Y = (L - \text{VaR}_\alpha)_+$ , the excess of the loss over VaR (0 below it)
  - ▶ Influence function  $\text{IF}(x) = \text{VaR}_\alpha + (x - \text{VaR}_\alpha)_+ / \alpha - \text{ES}_\alpha$  [Chen \(2008\)](#);  $\text{Var}(\widehat{\text{ES}}_\alpha) \approx \text{Var}(\text{IF}(L)) / n$
- ▣ **Tasks**
  1. Show that  $\text{Var}(\text{IF}(L)) = \text{Var}(Y) / \alpha^2$  and explain, with A7, why estimating VaR adds no first-order term.
  2. For  $L \sim N(0, 1)$ ,  $\alpha = 2.5\%$ , integrate, with  $v = \text{VaR}_\alpha = -z_\alpha$ ,  $\mathbb{E}[Y] = \int_v^\infty (x - v)\varphi(x) dx$  and  $\mathbb{E}[Y^2]$  in closed form.
  3. Compute the SE of  $\widehat{\text{ES}}_{2.5\%}$  for  $n = 500$  and  $\sigma = 1.2\%$ , and check it by simulation.
- ▣ **Report:** the two moments, the SE in pp and the simulated standard deviation
- ▣ **Interpretation:** why is the S&P 500 value (B6) much larger?
- ▣ **Worked example to follow:** A1, A7

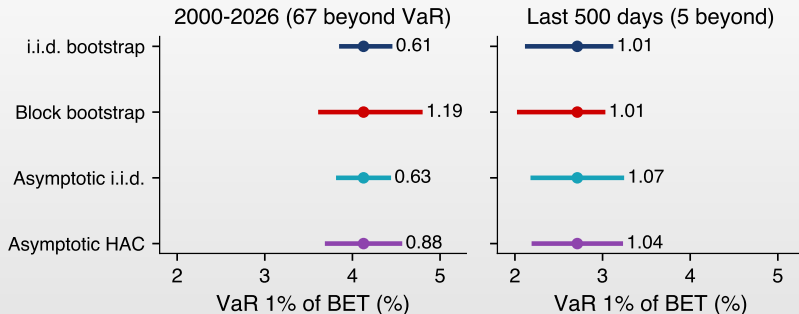
## B1 Extended [Solved]: Asymptotic and HAC Intervals for VaR — Task

- ▣ **Question:** do the asymptotic formulas agree with the bootstrap of B1?
- ▣ **Data**
  - ▶ i.i.d.:  $SE = \sqrt{\alpha(1-\alpha)/n} / \hat{f}(\widehat{VaR})$ ,  $\hat{f}$  = Gaussian kernel density of  $L$  (Silverman bandwidth)
  - ▶ HAC: replace  $\alpha(1-\alpha)$  by the Newey–West long-run variance of  $l_t = 1\{L_t > \widehat{VaR}\}$  (Bartlett kernel, lag  $\lfloor 4(n/100)^{2/9} \rfloor$ )
- ▣ **Tasks**
  1. Normal benchmark (from A1): with  $f(VaR_{1\%}) = \varphi(z_{1\%})/1.2$ , compute the SE for  $n = 500$  and  $n = 6\,670$ .
  2. For BET, compute  $\hat{f}$ , the i.i.d. and HAC SEs and the two 95% intervals on both samples.
- ▣ **Report:** the four intervals of VaR 1% next to the two bootstrap intervals of B1
- ▣ **Interpretation:** why are the BET intervals wider than the Normal benchmark?

## B1 Extended [Solved]: Solution

- ▣ 1.  $f = 0.0267/1.2 = 0.0222$ : SE 0.20 pp ( $n = 500$ , width 0.79), 0.055 pp ( $n = 6\,670$ , width 0.22)
- ▣ 2. BET full:  $\hat{f} = 0.0076$ , SE 0.16 (i.i.d.), 0.23 (HAC); intervals  $[3.81, 4.44]$ ,  $[3.69, 4.57]$
- ▣ Last 500 days:  $\hat{f} = 0.0163$ , SE 0.27 and 0.27
- ▣ Interpretation: the BET tail is heavier than the Normal one, so the density at VaR is lower and the quantile less precise; HAC and blocks agree that clustering widens the full-sample interval

## B1 Extended [Solved]: Solution — Chart



Dot: historical VaR 1%; bar: 95% interval; number: interval width in percentage points

## B3 Extended [Proposed]: A Bootstrap Interval for Tomorrow's FHS VaR — Task

- ▣ **Question:** how uncertain is the filtered VaR of B3?
- ▣ **Data:** the B3 model and data; residual bootstrap [Christoffersen & Gonçalves \(2005\)](#) with 999 paths, seed 42, function `fhs_bootstrap` of the seminar Quantlet
- ▣ **Tasks**
  1. For each path, simulate an AR(1)-GARCH(1,1) series of the sample length with resampled  $\hat{z}$  and re-estimate  $\theta^* = (c^*, \phi^*, \omega^*, \alpha_1^*, \beta_1^*)$ .
  2. Compute  $\mu_{t+1}^*$  and  $\sigma_{t+1}^*$  by running the **original** Bitcoin returns through  $\theta^*$ , and  $\hat{\xi}_{1\%}^*$  from the path's own residuals.
  3. Report the 90% percentile interval of  $\text{VaR}^* = -\mu_{t+1}^* + \sigma_{t+1}^* \hat{\xi}_{1\%}^*$  and the intervals of its two sources.
- ▣ **Report:** the histogram of the 999 forecasts with the interval
- ▣ **Interpretation**
  - ▶ does the interval include the 500-day historical VaR?
  - ▶ is that a test?
- ▣ **Worked example to follow:** B3

## B4 [Proposed]: Cornish–Fisher vs Empirical Quantiles — Task

- ▣ **Question:** does the Cornish–Fisher correction fix the Normal VaR on real data?
- ▣ **Data**
  - ▶ BET (compulsory): the B1 data; extension: S&P 500, Bitcoin and gold (daily closes, weekdays for gold) and the BNR EUR/RON reference rate (2005–2026), all to 18 September 2026
  - ▶ Moments of the returns  $X$ : mean, standard deviation with `ddof=1`, `scipy.stats.skew` and `scipy.stats.kurtosis` (excess, biased)
- ▣ **Tasks**
  1. Compute  $S$  and  $K$  for BET.
  2. Compute VaR 1% and VaR 2.5% by the Normal distribution, Cornish–Fisher (A6) and historical simulation.
  3. Build 95% i.i.d. bootstrap intervals (1 000 resamples) for the two historical VaRs and check whether the Cornish–Fisher values fall inside.
  4. Separately, check whether  $K$  exceeds the monotonicity bound of A6.
- ▣ **Report:** a small table per asset and a chart of the three methods with the bootstrap intervals
- ▣ **Interpretation:** why does Cornish–Fisher work better at 2.5% than at 1%?
- ▣ **Worked example to follow:** A1, A6, B1

## B5 Extended [Proposed]: Risk-Parity Weights — Task

- ▣ **Question:** which long-only, fully invested weights give each asset of B5 the same share of Normal VaR?
- ▣ **Data:** the B5 returns and covariance  $\Sigma$
- ▣ **Tasks**
  1. Solve  $w_i(\Sigma w)_i = w_j(\Sigma w)_j$  for all  $i, j$ , with  $w \geq 0$  and  $\sum_i w_i = 1$ , numerically.
  2. Compare with the inverse-volatility weights  $w_i \propto 1/\sigma_i$ .
  3. At the risk-parity weights, compute the Normal VaR 1%, the historical ES 2.5% and the ES shares.
- ▣ **Report:** the weights, VaR, ES and shares
- ▣ **Interpretation:** does equal VaR risk mean equal ES risk?
- ▣ **Worked example to follow:** B5

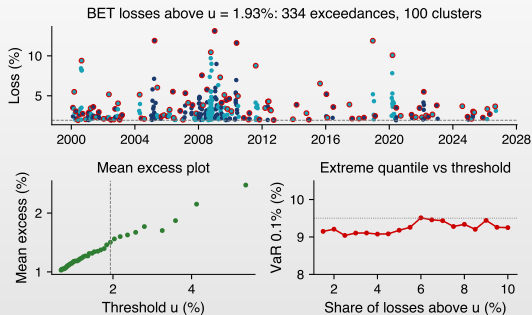
## B2 Extended [Solved]: Clustered Exceedances and the Extremal Index — Task

- ▣ **Question:** how many independent pieces of tail information do the 334 BET exceedances of B2 contain?
- ▣ **Data**
  - ▶ Exceedance times  $S_1 < \dots < S_N$  of  $u$  (B2), gaps  $T_i = S_{i+1} - S_i$
  - ▶ Intervals estimator [Ferro & Segers \(2003\)](#):  $\hat{\theta} = \min\left\{1, \frac{2(\sum_i (T_i - 1))^2}{(N-1) \sum_i (T_i - 1)(T_i - 2)}\right\}$  if  $\max T_i > 2$
- ▣ **Tasks**
  1. Compute  $\hat{\theta}$  and the target number of clusters  $C = \lfloor \hat{\theta}N \rfloor + 1$ .
  2. Split the exceedances at the  $C - 1$  largest gaps; if the  $(C - 1)$ -th and  $C$ -th largest gaps are equal, reduce  $C$  until they differ.
  3. Fit the GPD to the cluster maxima and compare  $\hat{\xi}/\text{SE}$  with B2.
- ▣ **Report:**  $\hat{\theta}$ , the number of clusters, the declustered fit and a timeline chart
- ▣ **Interpretation:** is the i.i.d. SE of B2 credible?

## B2 Extended [Solved]: Solution

- ▣ 1.  $\hat{\theta} = 0.32$ ,  $C = 108$
- ▣ 2. Tie at the cut ( $T = 12$  days): 100 clusters of 3.3 exceedances on average; largest cluster 33, 47 isolated exceedances
- ▣ 3.  $\hat{\xi} = 0.269$  (SE 0.127) vs 0.209 (SE 0.066);  $\hat{\xi}/\text{SE}$  falls from 3.2 to 2.1; at the VaR 1% threshold  $\hat{\theta} = 0.40$
- ▣ Interpretation: the tail stays heavy, but the i.i.d. SE is not credible; report the declustered fit or filter with GARCH first (conditional EVT)

## B2 Extended [Solved]: Solution — Chart



- Exceedances (clusters alternate colour)
- Cluster maximum
- Empirical mean excess
- GPD VaR 0.1%

## B7 [Proposed]: CAViaR on BET — Task

□ **Question:** does modelling the BET quantile directly beat filtered and historical simulation out of sample?

□ **Data**

- ▶ BET index, daily closes 2000-01-05 – 2026-09-18, weekdays only;  $n = 6\,670$  daily log returns in %,  $r_t = 100(\ln P_t - \ln P_{t-1})$ , loss  $L_t = -r_t$
- ▶ SAV CAViaR [Engle & Manganelli \(2004\)](#):  $\text{VaR}_t = \beta_0 + \beta_1 \text{VaR}_{t-1} + \beta_2 |r_{t-1}|$ ,  $|\beta_1| < 1$  (persistence of VaR),  $\beta_2$ : reaction to yesterday's absolute return,  $\text{VaR}_1 = -\hat{q}_1\%$  of the first 300 returns; routines `caviar_path`, `caviar_fit` (seminar Quantlet):  $10^4$  starts in  $(0, 1)^3$ , seed 42, the best 10 refined

□ **Tasks**

1. Estimate  $\beta$  on 2000–2015 by minimising the pinball loss  $\sum_t (\alpha - 1\{r_t < -\text{VaR}_t\})(r_t + \text{VaR}_t)$  at  $\alpha = 1\%$ .
2. Compute one pinball-loss term step by step for the first out-of-sample day and check that the forecast for day  $t$  uses only  $r_{t-1}$ .
3. With  $\beta$  fixed, forecast VaR for 2016–2026 and compare exceedances and mean pinball loss with FHS (refitted each January) and rolling 500-day HS.
4. For 20 February – 30 April 2020 count exceedances and report the peak VaR of each method.

□ **Report:** the estimates, a comparison table and a chart with a COVID-19 zoom

□ **Interpretation:** can the difference be attributed to the model class when CAViaR is fitted once and FHS every year?

□ **Worked example to follow:** B3; lecture, CAViaR on the S&P 500

## C1 [Open]: ES Capital, BVB vs S&P 500 — Task (1/2)

- ▣ **Question:** which portfolio needs more ES capital per unit invested: Bucharest blue chips or the S&P 500?
- ▣ **Data**
  - ▶ BVB basket: Banca Transilvania, OMV Petrom, BRD–Groupe Société Générale, Transgaz, Romgaz, Nuclearelectrica, Electrica, Transelectrica (adjusted closes), equal weights, rebalanced daily; days on which no stock traded are removed
  - ▶ S&P 500 portfolio: SPY, the S&P 500 ETF (adjusted closes); both from 2016-09-20 to 18 September 2026; daily simple returns  $R_t$  in % in local currency
- ▣ **Tasks**
  1. Construction: compute the one-day historical ES 2.5% of each portfolio (loss  $-R_t$ ) and its tail count.
  2. Horizon: compute the 10-day ES 2.5% by  $\sqrt{10}$  scaling, from overlapping compounded losses  $100[1 - \prod_{i=1}^{10}(1 + R_{t+i}/100)]$ , and by filtered Monte Carlo (AR(1)-GARCH(1,1) on log returns, today's state, 100 000 paths, seed 42, each path converted to a simple loss).
- ▣ **Report:** continued on the next slide
- ▣ **Worked example to follow:** B1, B3; lecture, horizon and Basel sections

## C1 [Open]: ES Capital, BVB vs S&P 500 — Task (2/2)

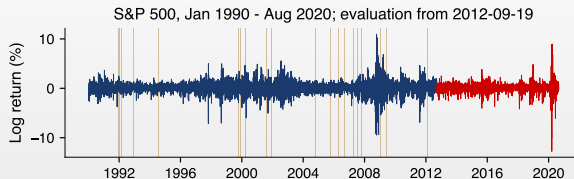
### ▣ Tasks (continued)

3. Stress: find the 250 consecutive days with the largest one-day historical ES 2.5%, plot the rolling ES with the window shaded, and report the stressed ES at 1 day and, by  $\sqrt{10}$ , at 10 days.
  4. Capital: capital per 1 000 000 invested =  $1\,000\,000 \times 1.5 \times \text{ES}_{10}^{\text{stress}} (\%) / 100$  (1.5: the FRTB multiplier  $m_c$ ;  $\text{ES}_{10}^{\text{stress}}$ : the 10-day stressed ES), for a 10-day and a 20-day liquidity horizon (a further factor  $\sqrt{2}$ ); amounts in RON for BVB and in USD for SPY, ratios comparable.
  5. Uncertainty: 95% moving-block bootstrap intervals (20-day blocks, 2 000 resamples) for the one-day ES and for the stressed ES with the window held fixed.
- ▣ **Report:** a table of all numbers with intervals and a one-page argument
- ▶ which portfolio needs more capital?
  - ▶ how do thin trading and currency change the answer?
- ▣ A simplified illustration inspired by FRTB [Basel Committee \(2019\)](#), not the full capital calculation of [Basel Committee, MAR33](#)

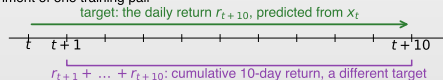
## C3 [Open]: Deep Quantile Regression for VaR — Task (1/2) — Minimum Version

- **Question:** does a small neural network beat linear quantile regression for the S&P 500 VaR 1%, as in the lecture case study [Chronopoulos, Raftapostolos & Kapetanios \(2024\)](#)?
- **Data**
  - ▶ S&P 500 daily closes, 1990-01-03 – 31 August 2020 (the paper starts in 1985); unscaled daily log returns  $r_t$ ; tuning: first 5 726 days, evaluation: last 2 000 (from 2012-09-19)
  - ▶ One predictor set: GARCH(1,1)  $\sigma_t$  (estimated on the tuning sample only);  $\alpha = 1\%$
- **Tasks**
  1. Build training pairs  $(x_t, r_{t+10})$  whose targets all lie in the tuning sample, as in the chart.
  2. Fit a linear quantile regression and one network (one hidden ReLU layer of 5 units, ridge penalty 0.01, Adam optimiser (stochastic gradient descent with adaptive step sizes), early stopping) by the pinball loss; standardise the predictors on the training part of each cross-validation split only.
  3. Compare mean pinball losses on the evaluation sample with a Diebold–Mariano test (HAC, 10 lags) [Diebold & Mariano \(1995\)](#), over 5 random seeds.
- **Report:** a table of mean pinball losses, DM statistics and exceedance counts, per seed

## C3 [Open]: Deep Quantile Regression for VaR — Data



Alignment of one training pair



— Tuning sample (5,726 days)

— Cross-validation: 20 random test days, each with  $\pm 5$  neighbouring days excluded

— Evaluation sample (last 2,000 days)

- Choose the target explicitly: the daily return  $r_{t+10}$ , not the cumulative 10-day return

## C3 [Open]: Deep Quantile Regression for VaR — Task (2/2) — Extensions

- ▣ **Replication extension:** the four predictor sets and the cross-validation grid of the paper (Appendix of the lecture); relative RMSFE only to reproduce Table 6, since RMSFE is minimised by the conditional mean, not the quantile
- ▣ **Rolling extension:** re-estimate once a year on a rolling window and forecast September 2020 – 2026 for the S&P 500 and the BET
- ▣ **Joint extension:** two outputs  $(v_t, e_t)$  trained on the FZ0 loss [Patton, Ziegel & Chen \(2019\)](#), ( $v_t$ : return quantile,  $e_t$ : tail mean, as in the lecture) with  $e_t = -\exp(a_t)$  and  $v_t = e_t + \exp(b_t)$ ,  $a_t, b_t$  unconstrained network outputs, so that  $e_t < 0$  and  $e_t \leq v_t$
- ▣ **Interpretation** (one page): can a gain be attributed to non-linearity when architecture, penalty and tuning change together?

## Project Choice and Common Rubric

### Choose one

- C1: ES capital for a portfolio of your choice (at least three assets not covered in class), with the four-step structure of C1
- C3: the minimum version, then one extension
- Deliver a Quantlet folder (code, `Metainfo.txt`, charts) and three slides: numbers, interpretation, limitations

### Rubric

- Reproducibility: data loaded by code, fixed seeds, every number from a script (30%)
- Correct conventions: sign, level, horizon, return type (20%)
- Uncertainty: tail counts and intervals for every tail estimate (25%)
- Interpretation and limits, including logged AI use and the AI errors actually found (25%)

## References (1/3)

- ▣ Acerbi, C., Tasche, D. (2002). [On the coherence of expected shortfall](#). *Journal of Banking & Finance*, 26(7), 1487–1503.
- ▣ Artzner, P., Delbaen, F., Eber, J.-M., Heath, D. (1999). [Coherent measures of risk](#). *Mathematical Finance*, 9(3), 203–228.
- ▣ Barone-Adesi, G., Giannopoulos, K., Vosper, L. (1999). [VaR without correlations for portfolios of derivative securities](#). *Journal of Futures Markets*, 19(5), 583–602.
- ▣ Basel Committee on Banking Supervision (2019). [Minimum capital requirements for market risk](#). Bank for International Settlements.
- ▣ Chen, S.X. (2008). [Nonparametric estimation of expected shortfall](#). *Journal of Financial Econometrics*, 6(1), 87–107.
- ▣ Christoffersen, P., Gonçalves, S. (2005). [Estimation risk in financial risk management](#). *Journal of Risk*, 7(3), 1–28.
- ▣ Chronopoulos, I., Raftapostolos, A., Kapetanios, G. (2024). [Forecasting Value-at-Risk using deep neural network quantile regression](#). *Journal of Financial Econometrics*, 22(3), 636–669.
- ▣ Cornish, E.A., Fisher, R.A. (1938). [Moments and cumulants in the specification of distributions](#). *Revue de l'Institut International de Statistique*, 5(4), 307–320.
- ▣ Diebold, F.X., Mariano, R.S. (1995). [Comparing predictive accuracy](#). *Journal of Business & Economic Statistics*, 13(3), 253–263.

## References (2/3)

- ▣ Efron, B. (1979). [Bootstrap methods: another look at the jackknife](#). *Annals of Statistics*, 7(1), 1–26.
- ▣ Engle, R.F., Manganelli, S. (2004). [CAViaR: conditional autoregressive value at risk by regression quantiles](#). *Journal of Business & Economic Statistics*, 22(4), 367–381.
- ▣ Ferro, C.A.T., Segers, J. (2003). [Inference for clusters of extreme values](#). *Journal of the Royal Statistical Society: Series B*, 65(2), 545–556.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). [Statistics of Financial Markets: An Introduction](#), 5th ed. Springer.
- ▣ Harvey, D., Leybourne, S., Newbold, P. (1997). [Testing the equality of prediction mean squared errors](#). *International Journal of Forecasting*, 13(2), 281–291.
- ▣ Künsch, H.R. (1989). [The jackknife and the bootstrap for general stationary observations](#). *Annals of Statistics*, 17(3), 1217–1241.
- ▣ McNeil, A.J., Frey, R. (2000). [Estimation of tail-related risk measures for heteroscedastic financial time series: an extreme value approach](#). *Journal of Empirical Finance*, 7(3–4), 271–300.
- ▣ McNeil, A.J., Frey, R., Embrechts, P. (2015). [Quantitative Risk Management: Concepts, Techniques and Tools](#), revised ed. Princeton University Press.
- ▣ Patton, A.J., Ziegel, J.F., Chen, R. (2019). [Dynamic semiparametric models for expected shortfall \(and Value-at-Risk\)](#). *Journal of Econometrics*, 211(2), 388–413.

## References (3/3)

- ▣ Pele, D.T., Mazurencu-Marinescu-Pele, M. (2026). Finite-sample precision limits for expected shortfall forecast comparisons. *Mathematics*, 14(13), 2316.
- ▣ Rockafellar, R.T., Uryasev, S. (2000). Optimization of conditional value-at-risk. *Journal of Risk*, 2(3), 21–41.
- ▣ Tasche, D. (2008). Capital allocation to business units and sub-portfolios: the Euler principle. arXiv:0708.2542.